

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 03: Tuesday 18 August 2026

(Week 2)

Remark. Q02.14 to Q02.19 are leftover exercises from the Friday class in Week 1.

Q02.14. Find $\int x \cos(x^2 + 1) dx$.

Q02.15. Find $\int \frac{e^{3x}}{\sqrt{2e^{3x} + 5}} dx$.

Q02.16. Find $\int x \ln x dx$.

Q02.17. Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Q02.18. Evaluate $\int_0^{\frac{\pi}{2}} x \cos x dx$.

Q02.19. Find $\frac{d}{dx} \int_1^{\sin x} \sqrt{1+t^4} dt$.

Q03.1. Evaluate the improper integral

$$\int_0^{\infty} \frac{1}{1+x^2} dx.$$

Q03.2. Evaluate the improper integral

$$\int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx.$$

Q03.3. Find the area of the region bounded by the curve $y = \ln(x+3)$, the curve $y = 2 \ln x$ and the x -axis.

Q03.4. (Disk Method) The region bounded by curve $y^2 = x$, the line $y = x - 2$ and the y -axis is rotated about the x -axis. Find the volume of the solid generated.

Q03.5. (Shell Method) The region bounded by the curve $y = \ln(x - 1)$, the x -axis, the y -axis and the line $y = \ln 2$ is revolved about the x -axis. Use the shell method to find the volume of the solid generated.

Remark. In an earlier version of Q03.5, the axis of rotation was omitted.

Q03.6. Find the length of the curve

$$y = \frac{x^2}{8} - \ln x, \quad 1 \leq x \leq 2.$$

Q03.7. (i) Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I , and take a point $a \in I$. Let $g(x) = b + c(x - a) + d(x - a)^2$ be a quadratic polynomial, where b, c, d are real constants. Suppose

$$g(a) = f(a), \quad g'(a) = f'(a) \quad \text{and} \quad g''(a) = f''(a).$$

Find b, c, d in terms of $f(a)$, $f'(a)$ and $f''(a)$.

(ii) Now let $a = 1$ and suppose $f(1) = 2$, $f'(1) = 3$ and $f''(1) = 5$. Write down $g(x)$ in this case. Then using $g(x)$ as an approximation of $f(x)$, find an approximate value of $f(1.05)$.

Q03.8. Using the Maclaurin series of e^x given by

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad -\infty < x < \infty,$$

find the Taylor series of the function $f(x) = e^{2x-2}$ at $x = 3$.

Q03.9. Consider the vectors $\vec{A} = \langle 1, 5, 6 \rangle$ and $\vec{B} = \langle 2, 1, -3 \rangle$.

(i) Find $2\vec{A} - \vec{B}$.

(ii) Find the lengths of $\vec{A}$ and $\vec{B}$.

(iii) Find the unit vector with the same direction as $\vec{B}$.

(iv) Find the dot product $\vec{A} \cdot \vec{B}$.

(v) Hence find the angle between $\vec{A}$ and $\vec{B}$. Give your answer in radians, correct to two decimal places.